
NSF FUTURE MANUFACTURING DATA CHALLENGE | FINALIST PRESENTATION

Predicting DED track geometry via multi-modal sensor fusion

Fusing in-situ thermal history with substrate surface awareness to predict local track height

Team Submission

July 31, 2026 | github.com/alphons3t/nsf-chuds

Can we predict part geometry during printing?

- **Directed Energy Deposition (DED)** builds metal parts track by track — but final geometry varies along every track.
- In-situ sensors watch the melt pool in real time; **post-process metrology** reveals what was actually built — hours later.
- The challenge: connect what the camera **sees during printing** to the **local geometric variation** measured after printing.
- If we can do this, we unlock **real-time quality prediction and closed-loop control** for metal additive manufacturing.

OUR TASK

Predict local track height along the laser path from in-situ monitoring data

Ground truth: leveled 3D profilometer height maps

Inputs: thermal video + SEM substrate imagery

Scope: 4 tracks spanning different laser powers & scan speeds

Three sensors, three languages: time, texture, and space

THERMAL VIDEO

In-situ melt pool dynamics

- 400 frames per track · 50 fps
- 14 μm /pixel · 5.6 × 5.6 mm field of view
- Captures melt pool length, width, area
- Lives in the TIME domain

SEM IMAGERY

Substrate surface morphology

- Tiled .tif images along each track
- Reveals gouges, texture & roughness
- Used only outside the track (no leakage)
- Lives in the TEXTURE domain

3D PROFILOMETRY

Post-process ground truth

- Bruker/Wyko full-field height maps
- x, y in mm · z resolved to nanometers
- Defines the true deposited track height
- Lives in the SPACE domain

The core difficulty: these modalities share no common coordinate system — dynamic video frames must be fused with static spatial maps before any model can learn.

From three raw sensors to one physical dataset

1

Level

Fit & subtract a 2D plane from the raw 3D scan to isolate true track height

**2**

Align

Map 400 thermal frames onto the 20–100 mm spatial axis of the track

**3**

Engineer

Build thermal-history and substrate-awareness features at each location

**4**

Predict

Train a Random Forest on 4 tracks to predict local track height

Why this matters: every downstream feature is expressed in physical millimeter coordinates along the track — the model learns physics, not sensor artifacts.

Generative AI (LLM) was used transparently as a collaborative coding tool: designing the leveling math, debugging dataframe alignment, and refining visualizations.

Building a common coordinate system

STEP 1 — MATHEMATICAL 3D LEVELING

Recover the true track from a tilted, warped substrate

- Raw profilometer scans carry significant **planar tilt**.
- We isolate the **bare substrate edges** (non-track regions) and fit a 2D plane via linear regression.
- Subtracting the fitted plane levels the substrate to ≈ 0 , exposing the **true deposited height** — our ground truth.

STEP 2 — SPATIAL–TEMPORAL ALIGNMENT

Fuse video time with physical space

- Melt pool **length, width & area** extracted from each thermal frame by intensity thresholding (value = 2000).
- Constant scan speed (10 mm/s @ 50 fps) → each frame advances **0.2 mm** along the track.
- Linear interpolation maps all 400 frames onto the **20–100 mm coordinates** of the leveled height profile.

Feature engineering: physical memory and context

THERMAL HISTORY

DED is a continuous thermal process — not independent frames

- **Rolling averages** of melt pool dimensions over 5- and 10-frame windows.
- These act as a proxy for **heat accumulation and cooling rate** at each point along the path.
- Captures the longitudinal variation a single frame cannot see.

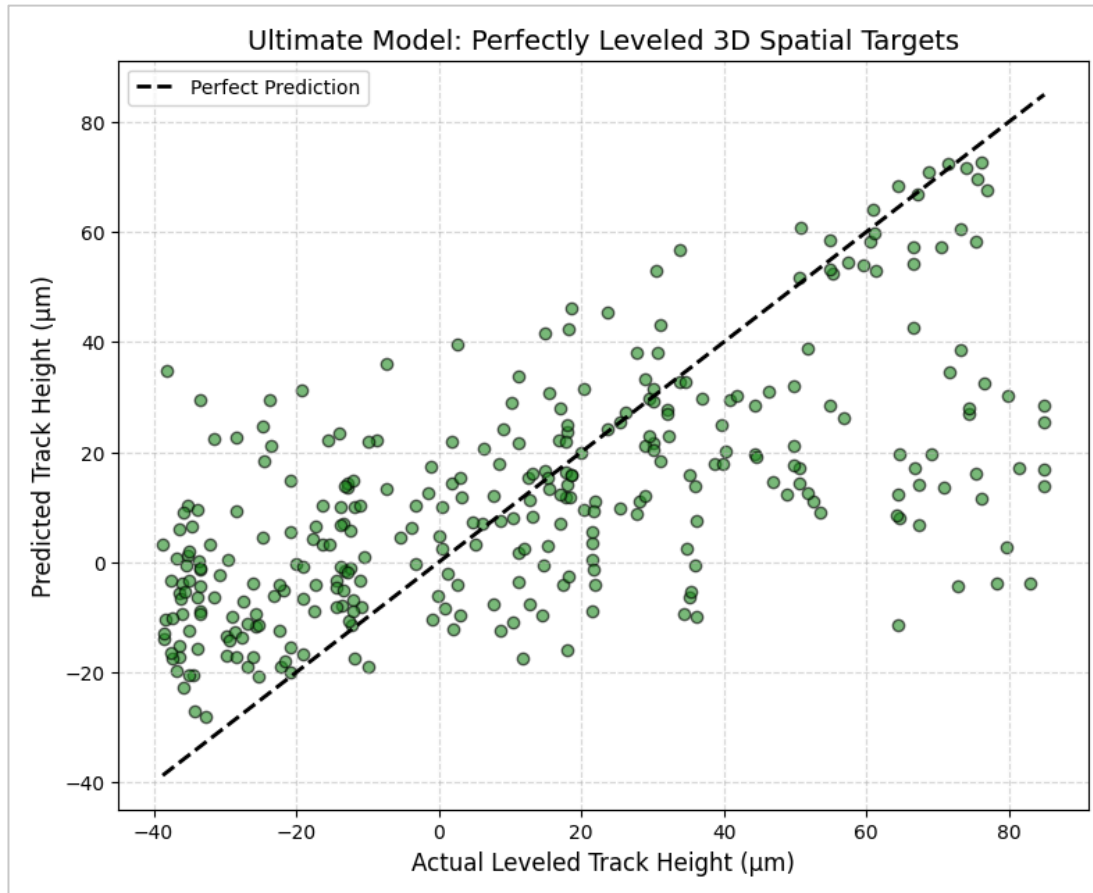
SUBSTRATE AWARENESS

The surface the laser lands on shapes what it builds

- Localized **SEM intensity and SEM roughness** matrices extracted from masked SEM imagery.
- Quantifies the **topographical foundation** prior to deposition — gouges, texture, reflectivity.
- Track region masked out to strictly avoid output leakage.

The model tracks real variation across four recipes

EXHIBIT 1 — ACTUAL VS PREDICTED TRACK HEIGHT



21.18 μm

Mean Absolute Error

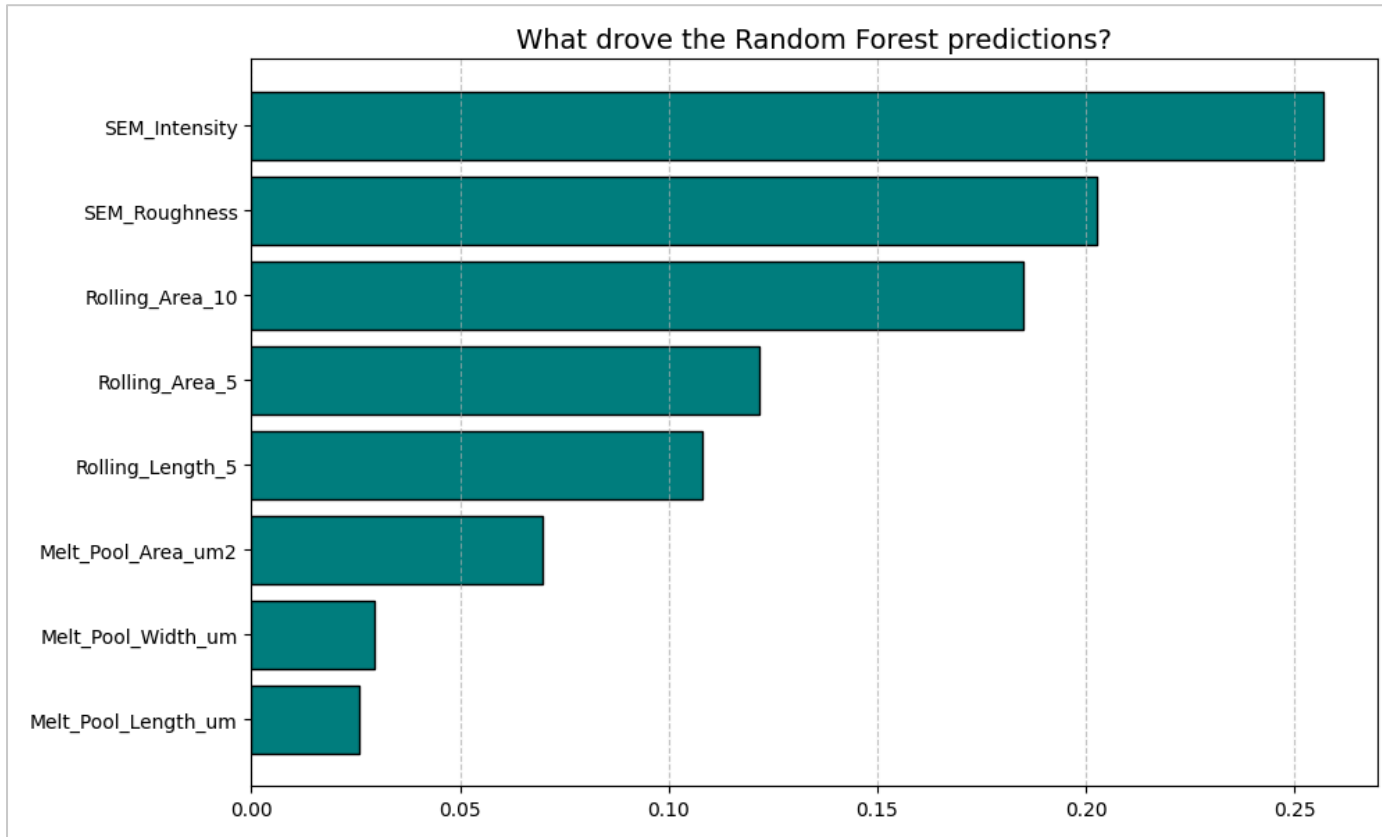
0.41 R^2

held-out test data

- **Random Forest Regressor** — 150 estimators, max depth 12, 80/20 train-test split.
- Trained jointly on **Tracks 8, 10, 14 & 21** to generalize across laser powers and scan speeds.
- Captures longitudinal variation via thermal rolling averages and constrains boundaries via melt pool width.
- Validated directly against **leveled profilometer ground truth** in physical μm .

The substrate — not the melt pool — drives prediction

EXHIBIT 2 — RANDOM FOREST FEATURE IMPORTANCE



WHAT THE FOREST LEARNED

SEM intensity is the single most dominant feature

1. Substrate texture governs heat dissipation and liquid metal flow.
2. Thermal history (rolling melt pool areas) ranks next — heat buildup matters.
3. Instantaneous melt pool size alone is the weakest signal.

Limitations & uncertainty: physics pushes back

- **Thermal warping:** the metal substrate bows during printing, so a perfectly rigid 2D plane fit cannot capture all deformation.
- **Deep gouges** in the substrate produce some negative ground-truth heights (down to $-40\text{ }\mu\text{m}$) after leveling.
- **Model uncertainty** grows in regions of extreme substrate deformation — exactly where the plane assumption weakens.
- $R^2 = 0.41$ reflects genuine stochasticity in DED, not just model error — a meaningful signal in a noisy physical process.

IF WE HAD MORE TIME

- Replace the rigid plane with a flexible spline / polynomial surface to absorb warping.
- Quantify prediction intervals per location (e.g., quantile forests).
- Extend from height to full width & boundary-position descriptors.
- Validate across additional tracks and unseen laser recipes.

Thermal monitoring alone is not enough

1

Fusion is the answer

Only by combining thermal history with substrate surface awareness can DED geometry be predicted reliably.

2

Physics-grounded pipeline

Automated 3D leveling + spatial-temporal alignment turned three incompatible sensors into one physical dataset.

3

Quantified & validated

MAE of 21.18 μm and R^2 of 0.41 against leveled profilometer ground truth, generalized over 4 laser recipes.

“The foundation a part is built on matters as much as the energy used to build it.”

Thank you — Questions? | github.com/alphons3t/nsf-chuds